MIRKO AMICO

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Education

Graduate Center, City University of New York, Aug. 2016 - Present
New York City College of Technology, CUNY, New York, U.S.A.
Ph.D. Candidate in Physics, Advisor: Roman Ya. Kezerashvili

Universita' degli Studi di Milano-Bicocca, Milan, Italy Sep. 2013 - Mar. 2016
M.S. in Physics (*Laurea Magistrale in Fisica*),
Advisor: Alessandro Tomasiello

Politecnico di Milano, Milan, Italy Sep. 2010 - Jul. 2013
B.S. in Physics Engineering (*Laurea in Ingegneria Fisica*)

Interests

Quantum information, Quantum computation, Superconducting circuits

Research

Research Assistant, City University of New York Aug. 2016 - Present
Ph.D. Dissertation: Tunable Quantum Entanglement in Superconducting Circuits.

- Investigate novel quantum vacuum phenomena that arise in a nonstationary cavity, such as the dynamical Casimir effect and the dynamical Lamb effect.
- Study the tunability of the quantum entanglement in a system of multiple qubits due to the nonstationary cavity.
- Propose physical realizations of a system of qubits coupled to a cavity with superconducting circuits

Research Project: Detector's Acceptance rate determination through Monte Carlo simulation of Muon's trajectories.

- Implemented a Monte Carlo simulation of 3D muons' trajectories in C.
- Determined the acceptance rate of a detector using the simulation of muons' trajectories.

Research Assistant, Universita' degli Studi di Milano-Bicocca Jan. 2015 - Mar. 2016
Master Thesis: IIA Linearized Equations of Motions on AdS_7xS_3 Background.

- Linearized the field equations of Type IIA supergravity using the formalism of generalized complex geometry.
- Expressed the linearized equations in terms of the fluctuations on a compactified AdS_7xS_3 manifold through the Kaluza-Klein decomposition.

Research Assistant, Deutsches Elektronen-Synchrotron Jul. 2014 - Sep. 2014
Research Project: Balanced Optical Cross-Correlator.

- Designed an optical setup for coherent synthesis between different laser beams by using a fraction of the combined beams.
- Built a Balanced Optical Cross-Correlator (BOC) for Ultrafast Optics to detect and compensate for timing jitter between different laser pulses.

Research Assistant, Politecnico di Milano May 2013 - Jul. 2013
Bachelor Thesis: Picosecond Acoustics.

- Investigated the opto-mechanical properties of a sample using the ultrafast optics technique of Picosecond Acoustics.

Teaching Adjunct, City University of New York Aug. 2016 - Present

- Teach college level lectures and experiments on physics.
- Worked as continuing education teacher for mathematics and physics.

Publications

[1] **M. Amico**, O. L. Berman, R. Ya. Kezerashvili, *Tunable quantum entanglement of three qubits in a non-stationary cavity*, Phys. Rev. A 96, 032328

[2] **M. Amico**, O. L. Berman, R. Ya. Kezerashvili, *Time evolution of the quantum entanglement between N qubits due to dynamical Lamb effect in the presence of dissipation*. Work in progress

[3] R. Ya. Kezerashvili, **M. Amico**, *Lectures on Quantum Computing* (Springer, New York). Work in progress.

Academic Activities

Poster presentation, "*Tuning the quantum entanglement of qubits in a nonstationary cavity*", New York City College of Technology, CUNY, Brooklyn — Nov. 2017

APS March Meeting presentation, "*Tuning the quantum entanglement of three qubits in a nonstationary cavity*", Los Angeles — Mar. 2018

Awards

CUNY Science Fellowship — 2016 - 2021

Doctoral Student Research Grant — 2018/2019

Early Research Initiative Provost's Pre-Dissertation Research Fellowship for the Sciences — 2018

Programming

C, Matlab, Python, QISKit